Telco Customer Churn Analysis

Project Overview

This project performs **Exploratory Data Analysis (EDA)** on a telecom dataset to identify the key factors driving customer churn.
The analysis focuses on uncovering **patterns, trends, and relationships** between customer attributes and churn behavior using visualizations and statistical insights.

Business Problem

Customer churn is a critical issue in the telecom industry, where acquiring new customers is more expensive than retaining existing ones.

👉 Objectives:

- Identify **high-risk customers likely to churn**
- Understand **major churn drivers**
- Provide **actionable business recommendations**

📂 Dataset Summary

- Total Customers: **7,043**
- Total Features: **21**
- Target Variable: **Churn (Yes/No)**

📊 Exploratory Data Analysis – Key Insights

🔹 1. Overall Churn Rate

- Churned Customers: **26.5%**
- Retained Customers: **73.5%**

👉 Approximately **1 in every 4 customers leaves**, indicating a significant retention problem.

🔹 2. Demographic Analysis

- Senior Citizens churn: **~42%**
- Non-Senior churn: **~23%**
- Customers without partner/dependents: **~30–35% churn**

👉 Customers with **family connections show stronger retention**, while seniors are more likely to churn.

🔹 3. Tenure (Customer Lifetime)

- 0–12 months: **~45–50% churn**
- 12–24 months: **~20–25% churn**
- 24 months: **~10–15% churn**

👉 **Early-stage customers are the most vulnerable**, with churn decreasing significantly as tenure increases.

🔹 4. Contract Type Impact

- Month-to-Month: **~43–45% churn**
- One-Year Contract: **~11–12% churn**
- Two-Year Contract: **~2–3% churn**

👉 Customers on long-term contracts are **up to 15x less likely to churn**.

🔹 5. Payment Method

- Electronic Check: **~45% churn**
- Mailed Check: **~20–25% churn**
- Bank Transfer / Credit Card (Auto-pay): **~15–20% churn**

👉 **Auto-payment users are significantly more stable**, while manual payment users churn more.

🔹 6. Services & Product Usage

Customers without value-added services show higher churn:

- No Online Security → **~40%+ churn**
- No Tech Support → **~38–40% churn**
- No Online Backup / Device Protection → Higher churn

👉 Customers using multiple services are **more engaged and less likely to leave**.

🔹 7. Internet Service Type

- Fiber Optic: **~40–42% churn**
- DSL: **~20–25% churn**
- No Internet: Lowest churn

👉 Despite being a premium service, **fiber users churn the most**, indicating possible dissatisfaction or pricing issues.

🔹 8. Monthly Charges Impact

- High Monthly Charges (>70): **~35–40% churn**
- Medium Charges: **~25–30% churn**
- Low Charges: **~15–20% churn**

👉 Clear indication that **price sensitivity influences churn behavior**.

📈 Key Visualizations

The project includes:

- Countplots with **Churn comparison (hue analysis)**
- Subplots for multiple service features
- Distribution plots for **tenure & monthly charges**
- Comparative bar charts showing churn trends

👉 These visualizations effectively highlight **correlations between features and churn**

🔍 Key Findings (Critical Insights)

🚨 High-Risk Customer Segments

- New customers (low tenure) → **~50% churn risk**
- Month-to-month users → **~45% churn**

- High monthly charges customers → **~40% churn**
- Customers without add-on services → **~35–40% churn**

✅ Strong Retention Indicators

- Long-term contracts → **<5% churn**
- Auto-payment users → **~15% churn**
- Customers with bundled services → significantly lower churn

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⚠️ Major Problem Area

- **Fiber optic customers (~42% churn)** despite premium pricing

👉 Suggests issues with:

- Service quality
- Pricing strategy
- Customer expectations

🚀 Business Recommendations

1. Improve Early Customer Retention

- Focus on first **6–12 months**
- Provide onboarding offers and engagement programs

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2. Promote Long-Term Contracts

- Offer **discounts on yearly plans**
- Incentivize contract upgrades

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3. Encourage Service Bundling

- Combine services like:
    - Security + Backup + Tech Support
- Increase customer dependency and engagement

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4. Optimize Pricing Strategy

- Review high-cost plans
- Provide flexible or customized pricing options

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5. Improve Fiber Service Experience

- Analyze complaints and service gaps
- Align pricing with perceived value

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6. Promote Automatic Payments

- Offer cashback/discounts for auto-pay users
- Reduce churn from manual payment customers

🛠️ Tech Stack

- Python (Pandas, NumPy)
- Data Visualization (Matplotlib, Seaborn)
- Jupyter Notebook

📌 Conclusion

The analysis clearly shows that customer churn is driven by a combination of:

- **Customer tenure**
- **Contract type**
- **Pricing (monthly charges)**
- **Service usage behavior**

👉 By targeting high-risk segments and improving service value, telecom companies can significantly **reduce churn and increase customer lifetime value (CLV)**.